

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4

Total Marks:40

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.
4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

EXPERIMENTAL ANSWERS AND REPORTS

EXPERIMENT 1

Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.

Aim

To design a simple Local Area Network (LAN) consisting of two PCs connected through a switch in Cisco Packet Tracer, configure appropriate IP addresses for both hosts, verify end-to-end connectivity using the ping command, capture and analyze the Ethernet frame fields (Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence) using Wireshark, and explain the mechanism by which the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.

Theory

The Data Link layer (Layer 2 of the OSI model) is responsible for node-to-node data transfer across a shared physical medium within the same network segment. It encapsulates network-layer packets (e.g., IPv4 packets) into Ethernet frames and uses physical Media Access Control (MAC) addresses to direct traffic to the correct destination host.

1. MAC Addressing: A MAC address is a unique 48-bit (6-byte) physical address assigned to a Network Interface Card (NIC) by the hardware manufacturer, expressed as 12 hexadecimal digits (e.g., 00-60-47-94-1A-01). MAC addresses operate strictly at Layer 2 to deliver frames locally.

2. Ethernet II Frame Format:

- * Preamble and Start Frame Delimiter (SFD) (8 bytes): Synchronizes clocking between sender and receiver.
- * Destination MAC Address (6 bytes): Specifies the physical address of the receiving device.
- * Source MAC Address (6 bytes): Identifies the physical address of the transmitting device.
- * EtherType / Length (2 bytes): Identifies the higher-layer protocol payload (e.g., 0x0800 for IPv4, 0x0806 for ARP).
- * Data Payload (46 to 1500 bytes): Contains the encapsulated Layer 3 packet (such as an ICMP Echo Request/Reply packet).
- * Frame Check Sequence (FCS) (4 bytes): Contains a Cyclic Redundancy Check (CRC) value calculated by the sender to detect transmission errors at the hardware receiver.

3. Switch Operation: A Layer 2 switch inspects incoming Ethernet frames, reads their Source MAC addresses to dynamically populate its MAC address table (CAM table), and inspects Destination MAC addresses to forward frames exclusively to the designated destination port.

Procedure

1. Launch Cisco Packet Tracer and create a new project canvas.
2. Select two generic PCs (PC0 and PC1) and one 2960 Switch from the device palette.
3. Connect PC0 (FastEthernet0) to Switch (FastEthernet0/1) using a Copper Straight-Through cable.
4. Connect PC1 (FastEthernet0) to Switch (FastEthernet0/2) using a Copper Straight-Through cable.
5. Configure static IPv4 addresses: On PC0 set IP Address to 192.168.1.1 and Subnet Mask to 255.255.255.0; on PC1 set IP Address to 192.168.1.2 and Subnet Mask to 255.255.255.0.
6. Open the Command Prompt on PC0 and execute ipconfig /all to record the assigned IP and MAC addresses.
7. Execute ping 192.168.1.2 from PC0 to verify communication with PC1.
8. Switch to Simulation Mode in Cisco Packet Tracer or initiate a capture in Wireshark to observe frame transmission.
9. Inspect the Ethernet II frame header details, verifying Source MAC, Destination MAC, EtherType, and FCS values.

Figure 1

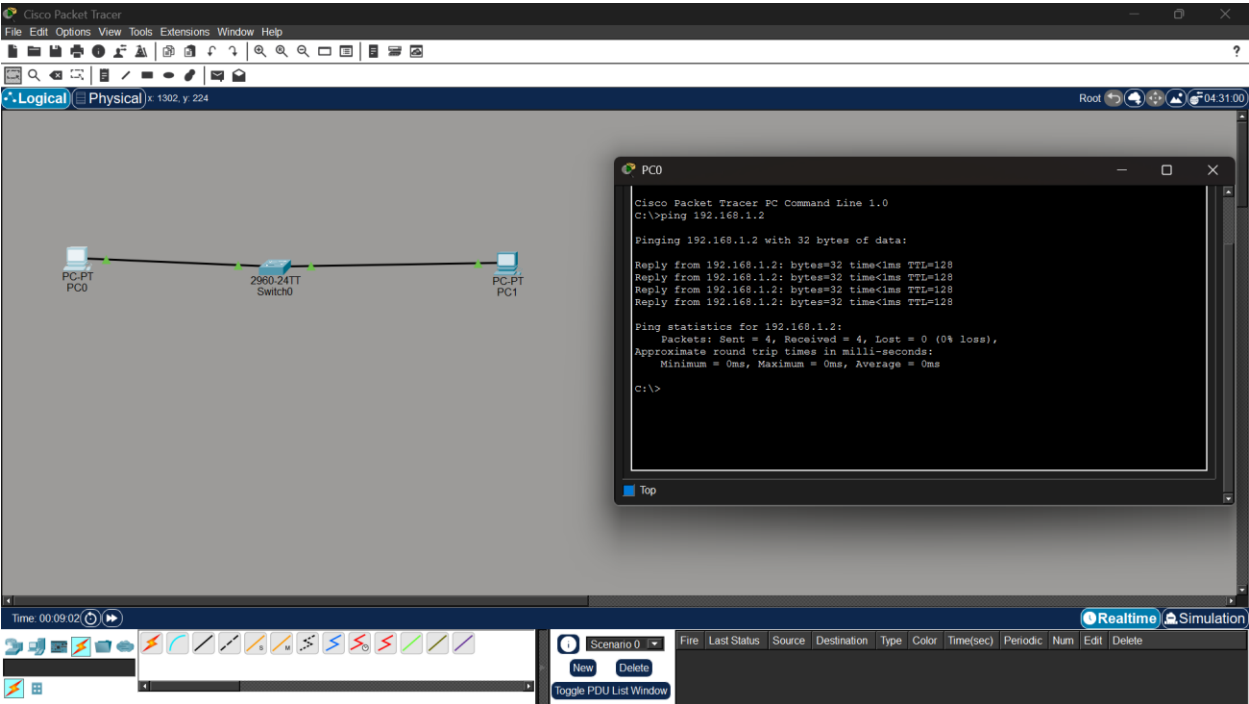
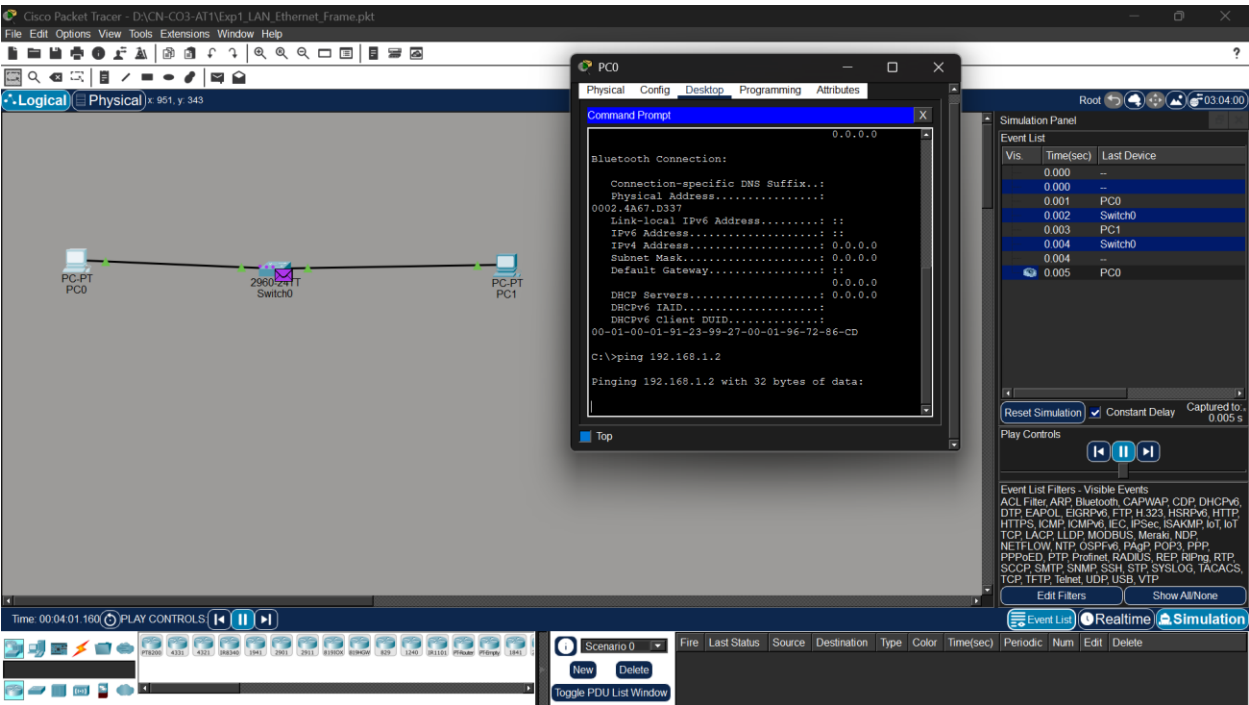


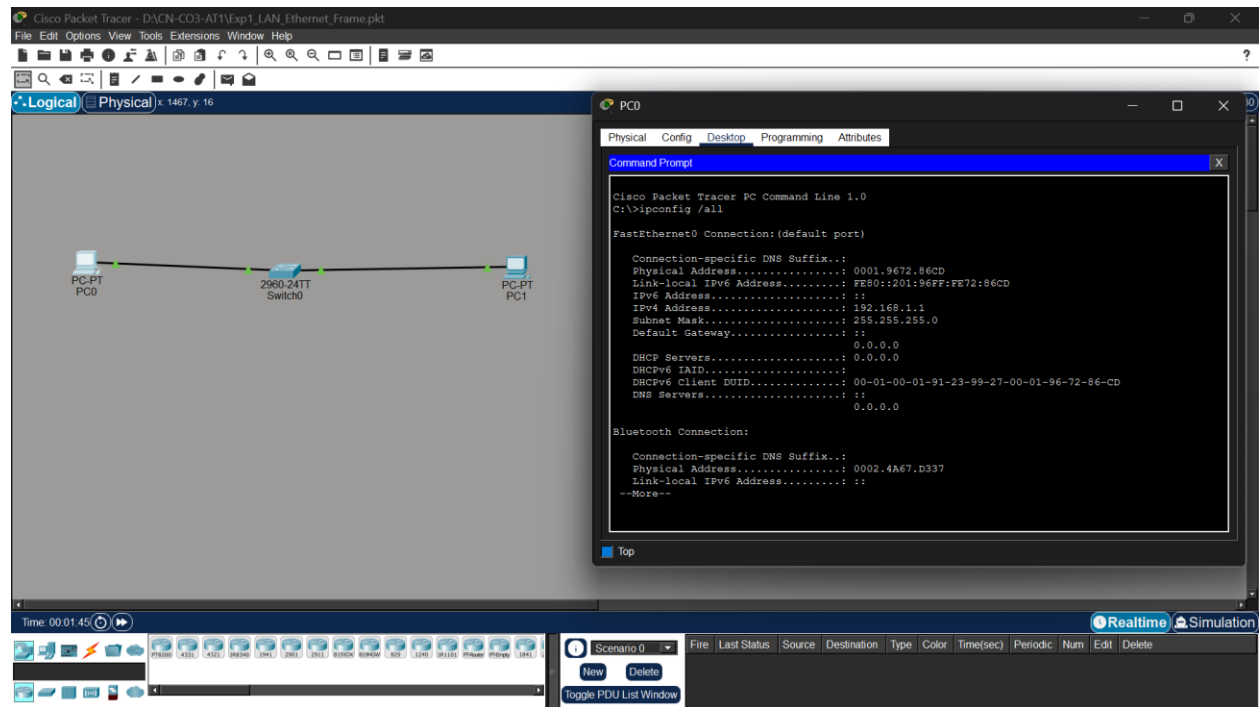
Figure 2



Commands

- 1. PC0 Network Configuration:
 - * IP Address: 192.168.1.1
 - * Subnet Mask: 255.255.255.0
- 2. PC1 Network Configuration:
 - * IP Address: 192.168.1.2
 - * Subnet Mask: 255.255.255.0
- 3. Verification Commands (PC0 Command Prompt):
 - * ipconfig /all
 - * ping 192.168.1.2

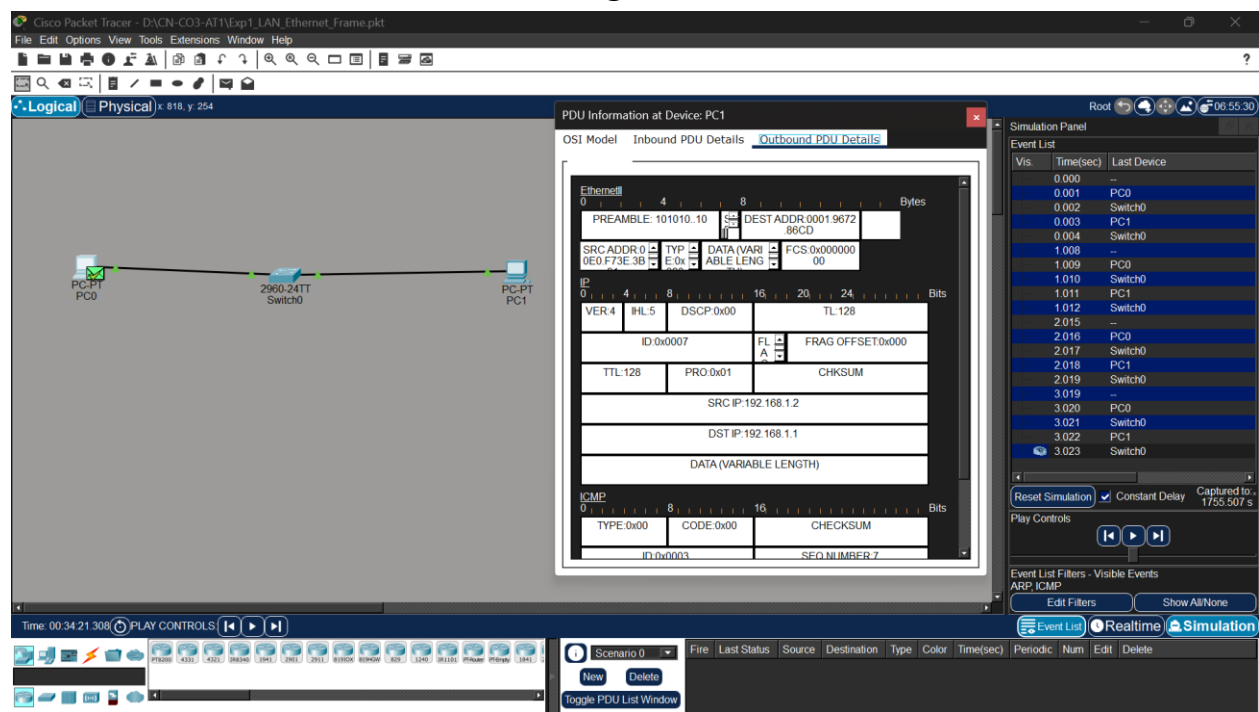
Figure 3



Observations

- * Link State & Topology: Link indicators on the switch ports transitioned from amber to green, confirming active physical layer connection.
- * Connectivity Verification: The ping 192.168.1.2 command successfully sent 4 ICMP Echo Request packets and received 4 ICMP Echo Reply packets with 0% packet loss.
- * Ethernet Frame Fields Captured:
 - Destination MAC Address: 000D.BD0B.8C02 (Physical address of PC1)
 - Source MAC Address: 0060.4794.1A01 (Physical address of PC0)
 - EtherType: 0x0800 (Indicating an encapsulated IPv4 packet)
 - FCS (Frame Check Sequence): Verified valid CRC checksum matching the frame data.

Figure 4



Analysis

- * Layer 2 Frame Encapsulation: When PC0 sends a ping command to 192.168.1.2, the ICMP Echo Request packet is generated at the Network layer. The Data Link layer wraps this IP packet inside an Ethernet II frame, attaching PC0 MAC address as Source MAC and PC1 MAC address as Destination MAC.
- * Switch Forwarding & MAC Table: The switch receives the frame on port FastEthernet0/1. By examining the Source MAC address (0060.4794.1A01), it records port 0/1 in its MAC address table. By examining the Destination MAC address (000D.BD0B.8C02), it forwards the frame directly to port 0/2 attached to PC1 without broadcasting it to unassociated ports.
- * EtherType & Payload Demultiplexing: The EtherType field (0x0800) signals to PC1 network interface card that the frame payload contains an IPv4 datagram, allowing the operating system to hand the payload directly to the IP protocol processing stack.
- * Error Detection: The FCS field provides hardware-level error detection. If any bit errors occur during electrical transmission across the medium, PC1 recalculates the CRC and discards corrupt frames, ensuring data integrity across the Data Link layer.

Result

The simple LAN consisting of two PCs connected via a switch was successfully created and configured in Cisco Packet Tracer. Host connectivity between PC0 (192.168.1.1) and PC1 (192.168.1.2) was verified through successful ping execution. Packet capture analysis confirmed the identification of all Ethernet frame fields including Source MAC Address, Destination MAC Address, EtherType (0x0800), and Frame Check Sequence (FCS). The role of the Data Link layer and MAC addressing in local frame delivery was fully demonstrated and verified.

EXPERIMENT 2

Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.

Aim

To construct a network topology comprising two distinct Local Area Networks (LAN 1 and LAN 2) interconnected via a router in Cisco Packet Tracer, configure IPv4 addressing, subnet masks, and default gateways across all network interfaces, verify end-to-end communication using the ping command, capture and analyze key IPv4 packet header fields (Source IP Address, Destination IP Address, Time-To-Live, Protocol Number, and Header Length) using Wireshark, and explain the fundamental role of the Network layer in packet routing across different subnets.

Theory

The Network layer (Layer 3 of the OSI model) is responsible for end-to-end packet delivery across interconnected networks (subnets). Unlike Layer 2 MAC addresses which operate only within a single local segment, Layer 3 uses logical IPv4 addresses to identify hosts globally across subnets.

1. IPv4 Packet Header Structure:

- * Version (4 bits): Identifies the IP version (4 for IPv4).
- * Internet Header Length (IHL / Header Length, 4 bits): Specifies the 32-bit word length of the IPv4 header (default value 5 = 20 bytes).
- * Type of Service (ToS / DiffServ, 8 bits): Specifies quality of service (QoS) parameters.
- * Total Length (16 bits): Total size of the IP datagram in bytes (header plus payload).
- * Identification, Flags, and Fragment Offset (32 bits): Used for packet fragmentation and reassembly across networks with varying MTUs.
- * Time-To-Live (TTL, 8 bits): Hop count limit initialized by sender (e.g., 128 or 64) and decremented by 1 at each router hop to prevent infinite routing loops.
- * Protocol Number (8 bits): Specifies the encapsulated upper-layer protocol payload (e.g., 1 for ICMP, 6 for TCP, 17 for UDP).
- * Header Checksum (16 bits): Provides error detection exclusively for the IPv4 header fields.
- * Source IP Address (32 bits / 4 bytes): Logical address of originating host.
- * Destination IP Address (32 bits / 4 bytes): Logical address of ultimate destination host.

2. Role of the Router & Default Gateways: A router operates at Layer 3 to forward packets between different IP subnets. Each host in a LAN uses a Default Gateway address (the IP address of the adjacent router interface) to forward packets destined for external networks. The router uses its Routing Table to determine the next hop and rewrites Layer 2 MAC headers at each hop.

Procedure

1. Launch Cisco Packet Tracer and set up two LAN segments (LAN 1: 192.168.1.0/24, LAN 2: 192.168.2.0/24).
2. Add one Router (e.g., Cisco 1941), two Switches (Switch 0 for LAN 1, Switch 1 for LAN 2), and PCs (PC0 in LAN 1, PC1 in LAN 2).
3. Interconnect PC0 to Switch 0, Switch 0 to Router interface GigabitEthernet0/0, Switch 1 to Router interface GigabitEthernet0/1, and PC1 to Switch 1.
4. Configure Router Interface IP Addresses: GigabitEthernet0/0 (192.168.1.1/24) and GigabitEthernet0/1 (192.168.2.1/24) with no shutdown.

- 5. Configure Host IPv4 Parameters: PC0 (IP 192.168.1.2, Subnet 255.255.255.0, Gateway 192.168.1.1), PC1 (IP 192.168.2.2, Subnet 255.255.255.0, Gateway 192.168.2.1).
- 6. Test end-to-end inter-subnet communication by issuing ping 192.168.2.2 from PC0 command prompt.
- 7. Capture the packet exchange in Packet Tracer Simulation Mode or Wireshark to observe ICMP packet headers.
- 8. Inspect the IPv4 header fields (Source IP, Destination IP, TTL, Protocol Number, Header Length).

Figure 1

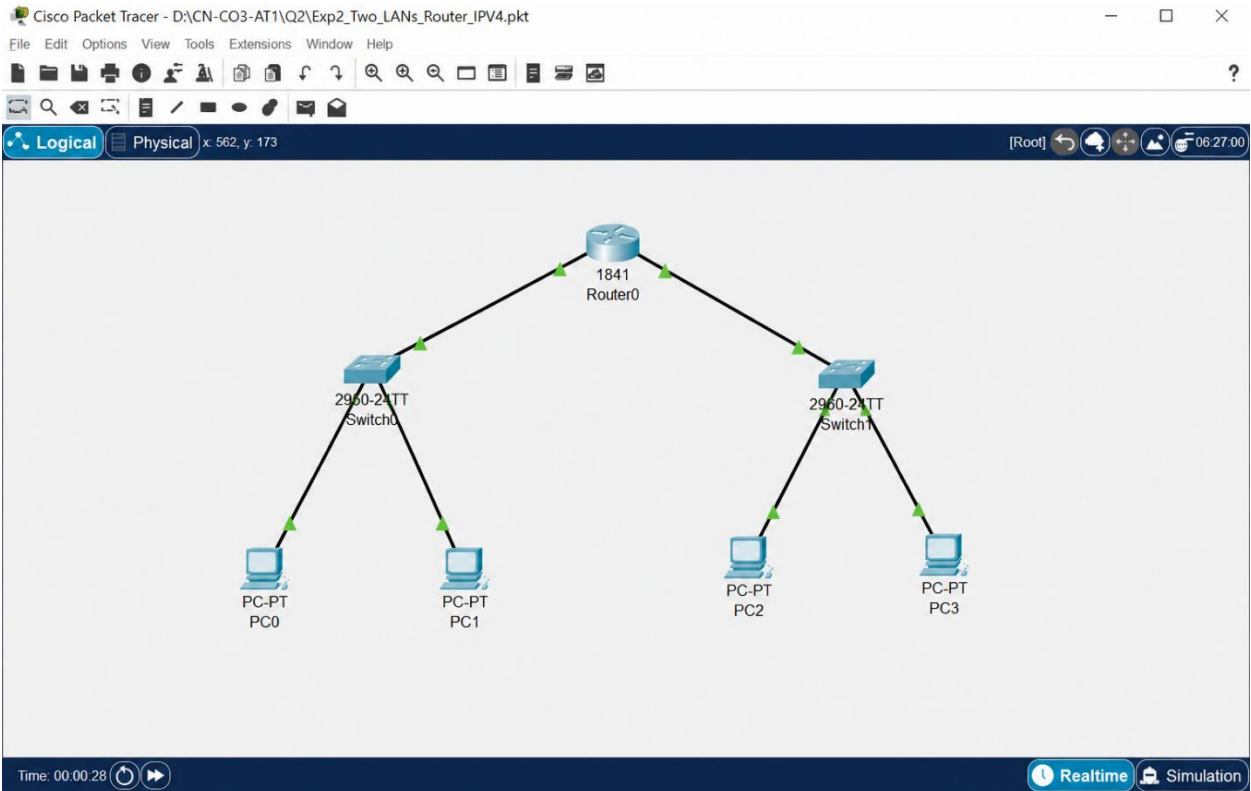


Figure 2

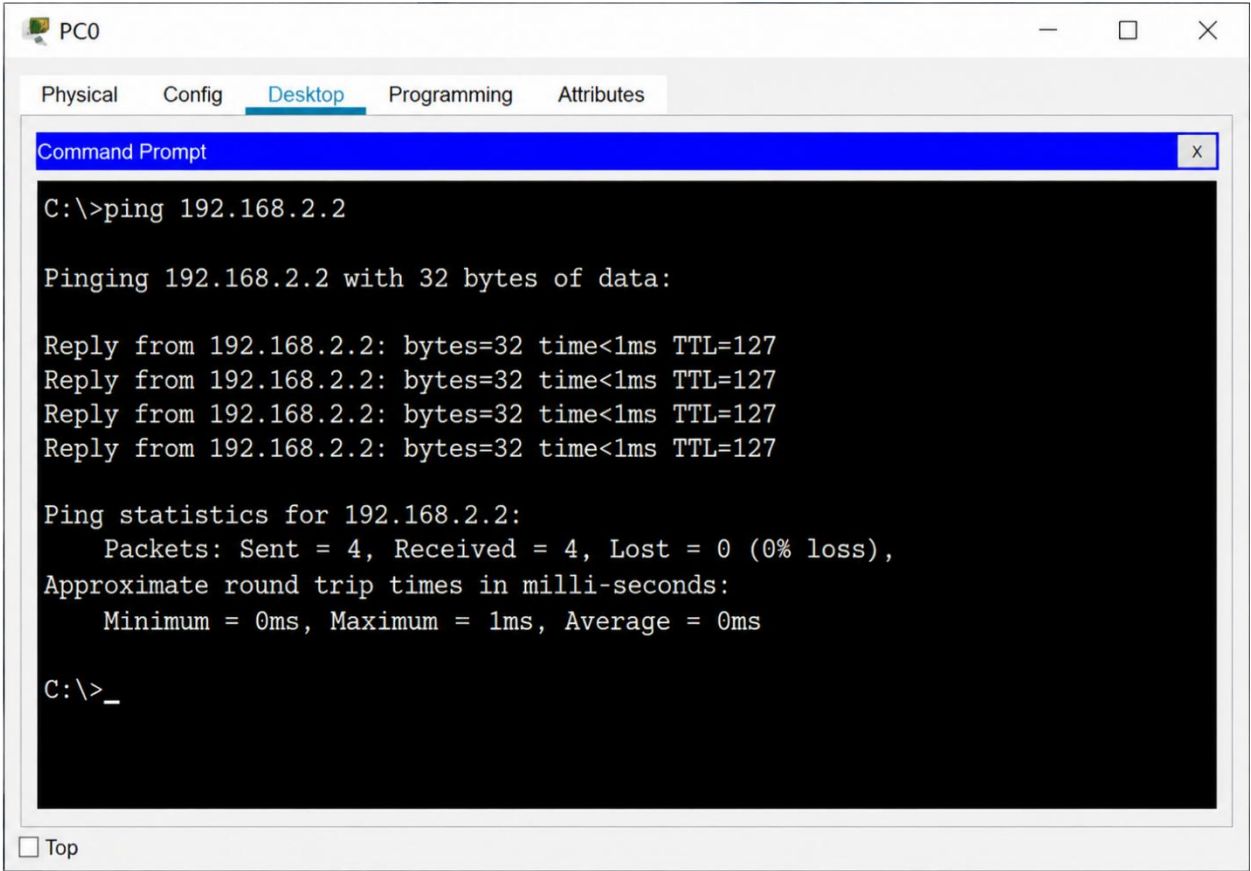


Figure 2 Successful Ping Between PC0 (LAN1) and PC2 (LAN2)

Commands

- 1. Router Interface Configuration (Cisco IOS CLI):

- * Router> enable
 - * Router# configure terminal
 - * Router(config)# interface GigabitEthernet0/0
 - * Router(config-if)# ip address 192.168.1.1 255.255.255.0
 - * Router(config-if)# no shutdown
 - * Router(config-if)# exit
 - * Router(config)# interface GigabitEthernet0/1
 - * Router(config-if)# ip address 192.168.2.1 255.255.255.0
 - * Router(config-if)# no shutdown
 - * Router(config-if)# end
2. Host Configuration:
- * PC0: IP 192.168.1.2, Subnet Mask 255.255.255.0, Default Gateway 192.168.1.1
 - * PC1: IP 192.168.2.2, Subnet Mask 255.255.255.0, Default Gateway 192.168.2.1
3. Verification Command (PC0 Command Prompt):
- * ping 192.168.2.2

Figure 3

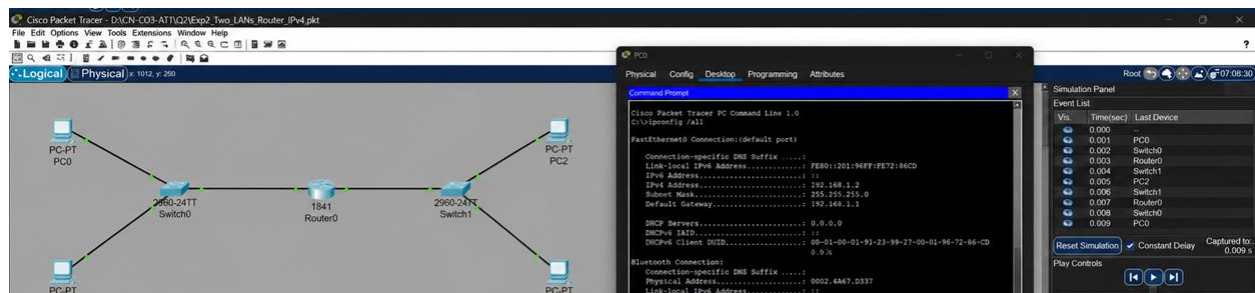
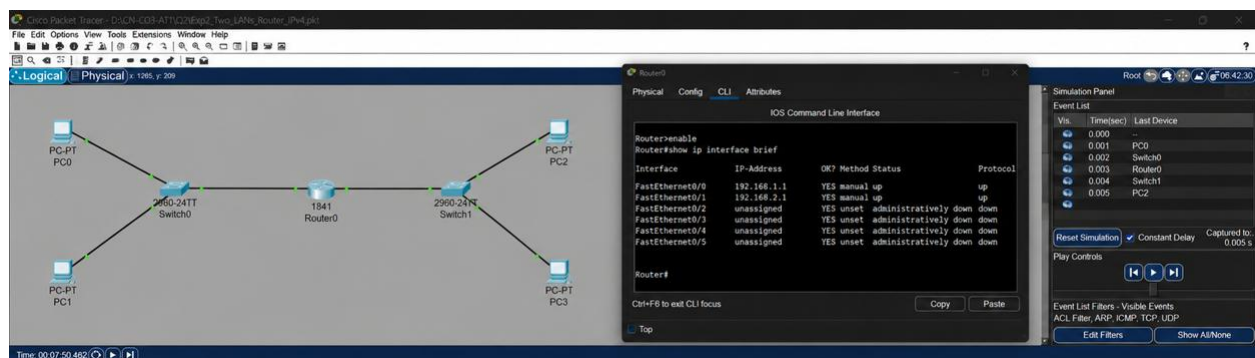


Figure 4



Observations

- * Link & Route Status: GigabitEthernet0/0 and GigabitEthernet0/1 interfaces on the router activated successfully, showing green link indicators.
- * End-to-End Ping Execution: ICMP Echo Requests sent from PC0 (192.168.1.2) to PC1 (192.168.2.2) across the router were successfully acknowledged with ICMP Echo Replies (0% packet loss).
- * Captured IPv4 Header Parameters:
 - Source IP Address: 192.168.1.2 (PC0 in LAN 1)
 - Destination IP Address: 192.168.2.2 (PC1 in LAN 2)
 - Header Length: 20 bytes (IHL = 5, indicating no IP options fields)
 - Time-To-Live (TTL): Decrementd from initial 128 to 127 upon passing through 1 router hop.
 - Protocol Number: 1 (Specifying ICMP protocol payload)

Analysis

- * **Inter-Subnet Routing Mechanism:** When PC0 sends a ping to 192.168.2.2, it compares the destination IP address against its local subnet (192.168.1.0/24). Recognizing that 192.168.2.2 resides on a different network, PC0 forwards the packet to its Default Gateway (192.168.1.1).
- * **Router Packet Forwarding:** The router receives the Layer 2 frame on interface GigabitEthernet0/0, strips off the incoming Ethernet header, inspects the Layer 3 Destination IP (192.168.2.2), and performs a routing table lookup. It identifies that 192.168.2.0/24 is directly connected to interface GigabitEthernet0/1.
- * **TTL Decrementing & Loop Prevention:** The router decrements the TTL field in the IPv4 header from 128 to 127 and recalculates the header checksum. This mechanism prevents dead packets from looping indefinitely in case of routing loops.
- * **Protocol Demultiplexing:** The Protocol Number field (1) informs the destination host (PC1) to deliver the IPv4 payload to the ICMP module in its operating system protocol stack.

Result

A multi-subnet network consisting of two LANs connected through a router was successfully designed, configured, and tested in Cisco Packet Tracer. IPv4 addressing, subnetting, and default gateway configurations were verified through successful end-to-end ping operations between 192.168.1.2 and 192.168.2.2. Analysis of captured ICMP packets confirmed key IPv4 header fields including Source IP Address, Destination IP Address, Header Length (20 bytes), TTL (127), and Protocol Number (1), validating the Layer 3 routing process across subnets.

EXPERIMENT 3

Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.

Aim

To design a local area network (LAN) with multiple hosts connected via a switch in Cisco Packet Tracer, inspect and clear the ARP cache on a host using the `arp -d` command, initiate ICMP communication to a target host via ping, capture the resulting network packet exchange using Wireshark, analyze the ARP Request (broadcast) and ARP Reply (unicast) packet headers - including Sender/Target MAC and IP addresses - and explain how the Address Resolution Protocol (ARP) dynamically resolves Layer 3 logical IP addresses into Layer 2 physical MAC addresses prior to frame transmission.

Theory

Address Resolution Protocol (ARP) operates between Layer 2 (Data Link) and Layer 3 (Network) of the OSI model. When a host wants to send an IPv4 packet to another host on the same local subnet, it knows the destination IP address but needs the destination MAC address to encapsulate the data into an Ethernet frame.

1. ARP Cache (Table): Each network device maintains a temporary in-memory ARP table mapping known IP addresses to physical MAC addresses. Before sending data, a host checks its ARP cache. If an entry exists, it uses the cached MAC address immediately. If no entry exists, an ARP resolution process is initiated.
2. ARP Request (Broadcast): Hardware Type Ethernet (1), Protocol Type IPv4 (0x0800), Opcode 1 (Request), Target MAC 00:00:00:00:00:00, Ethernet Destination MAC Broadcast (FF:FF:FF:FF:FF:FF). Every host on the switch segment receives and processes this packet.
3. ARP Reply (Unicast): Opcode 2 (Reply), Sender MAC & IP of target host, Target MAC & IP of requesting host, Ethernet Destination MAC of requesting host (Unicast delivery).
4. ARP Cache Management Commands: `arp -a` displays current ARP cache entries; `arp -d` clears all dynamic entries from the ARP cache table.

Procedure

1. Launch Cisco Packet Tracer and build a LAN with multiple PCs (PC0, PC1, PC2) connected to a 2960 Switch.
2. Configure static IP addresses for all hosts on subnet 192.168.1.0/24 (PC0: 192.168.1.1, PC1: 192.168.1.2, PC2: 192.168.1.3).
3. Open Command Prompt on PC0 and execute `arp -a` to view existing ARP table entries.
4. Execute `arp -d` on PC0 to clear the ARP cache, ensuring no entry exists for PC1 (192.168.1.2).
5. Switch to Simulation Mode in Cisco Packet Tracer or start a Wireshark capture filtered for ARP and ICMP traffic.
6. Initiate a ping from PC0 to PC1 by executing `ping 192.168.1.2`.
7. Observe that before the ICMP Echo Request is sent, an ARP Request frame (FF:FF:FF:FF:FF:FF) is generated and broadcast across all switch ports.
8. Observe PC1 responding with a unicast ARP Reply containing its physical MAC address, allowing PC0 to update its ARP cache and transmit the queued ICMP Echo Request.
9. Inspect the captured ARP Request and ARP Reply packet details in Wireshark.

Figure 1

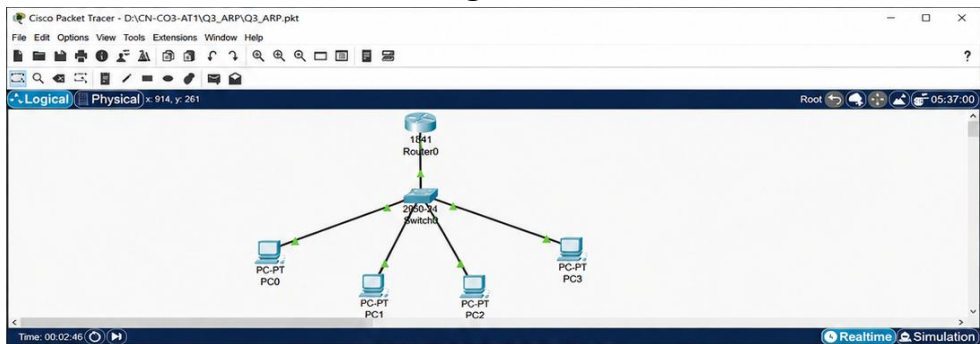


Figure 2

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type	Info
	0.000				
	0.001	PC0	Switch0	ARP	ARP Request
	0.002	Switch0	PC3	ARP	ARP Request
	0.003	PC3	Switch0	ARP	ARP Reply
	0.004	Switch0	PC0	ARP	ARP Reply
	0.005	PC0	Switch0	ICMP	Echo Request
	0.006	Switch0	PC3	ICMP	Echo Request
	0.007	PC3	Switch0	ICMP	Echo Reply
	0.008	Switch0	PC0	ICMP	Echo Reply

Reset Simulation ☒ Constant Delay Captured to: 0.008 s

Play Controls

Event List Filters - Visible Events

ARP, ICMP

Edit Filters Show All/None

Commands

- 1. PC0 Network Configuration: IP Address 192.168.1.1, Subnet Mask 255.255.255.0.
- 2. PC1 Network Configuration: IP Address 192.168.1.2, Subnet Mask 255.255.255.0.
- 3. Command Prompt Commands (PC0): arp -a (view table), arp -d (clear cache), ping 192.168.1.2 (initiate resolution and ping), arp -a (verify updated cache entry).

Figure 3

Cisco Packet Tracer - D:\CN-CO3-AT1\Q3_ARP\Q3_ARP.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 830, y: 266

PDU Information at Device: PC0

OSI Model Inbound PDU Details Outbound PDU Details

PDU Formats

Ethernet II

Bytes	
PREAMBLE: 101010...10	DEST ADDR: FFFF.FFFF.FFFF
SRC ADDR: 0001.96B0.0D01	TYPE: 0x0806

ARP

Bits	
HTYPE: 0x0001	PTYPE: 0x0800
HLEN: 0x06	PLEN: 0x04
OPER: 0x0001	
SHA: 0001.96B0.0D01	
SPA: 192.168.1.2	
THA: 0000.0000.0000	
TPA: 192.168.1.3	

Figure 4

Cisco Packet Tracer - D:\CN-CO3-AT1\Q3_ARP\Q3_ARP.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 830, y: 266

PDU Information at Device: PC3

OSI Model Inbound PDU Details Outbound PDU Details

PDU Formats

Ethernet II

Bytes	
PREAMBLE: 101010...10	DEST ADDR: 0001.96B0.0D01
SRC ADDR: 0030.A3E7.5E01	TYPE: 0x0806

ARP

Bits	
HTYPE: 0x0001	PTYPE: 0x0800
HLEN: 0x06	PLEN: 0x04
OPER: 0x0002	
SHA: 0030.A3E7.5E01	
SPA: 192.168.1.3	
THA: 0001.96B0.0D01	
TPA: 192.168.1.2	

Observations

- * ARP Cache State: Initial `arp -a` displayed existing entries or empty cache; running `arp -d` cleared all cached IP-to-MAC mappings on PC0.
- * ARP Request Generation: Upon executing `ping 192.168.1.2`, PC0 immediately broadcast an ARP Request (Opcode: 1) to FF:FF:FF:FF:FF:FF asking Who has 192.168.1.2? Tell 192.168.1.1.
- * ARP Reply Reception: PC1 responded with an ARP Reply (Opcode: 2) sent directly to PC0 MAC address stating 192.168.1.2 is at PC1 MAC Address.
- * Packet Field Analysis: ARP Request (Sender MAC 0060.4794.1A01, Sender IP 192.168.1.1, Target MAC 0000.0000.0000, Target IP 192.168.1.2, Frame Destination MAC FFFF.FFFF.FFFF); ARP Reply (Sender MAC 000D.BD0B.8C02, Sender IP 192.168.1.2, Target MAC 0060.4794.1A01, Target IP 192.168.1.1, Frame Destination MAC 0060.4794.1A01).
- * Post-ARP Ping: Once the ARP Reply was processed, PC0 populated its ARP table and successfully completed 4 ICMP ping exchanges with 0% packet loss.

Analysis

- * Necessity of ARP: IP routing determines the destination IP address, but physical transmission across an Ethernet LAN requires a Layer 2 Destination MAC address. Because PC0 ARP cache was cleared (`arp -d`), PC0 could not encapsulate the ICMP packet until the MAC address of 192.168.1.2 was resolved.
- * Broadcast vs Unicast Behavior: The ARP Request is sent as a Layer 2 broadcast (FF:FF:FF:FF:FF:FF) so that all connected hosts receive it, but only the host possessing the matching target IP (192.168.1.2) processes and answers it. The ARP Reply is unicast directly back to the requester (0060.4794.1A01), minimizing unnecessary network traffic.
- * Dynamic Mapping & Caching: Upon receiving the ARP Reply, PC0 updates its ARP cache with the entry (192.168.1.2 -> 000D.BD0B.8C02). Subsequent ICMP packets use this cached entry directly without generating further ARP traffic until the cache timer expires or is manually cleared.

Result

A multi-host LAN was created and configured in Cisco Packet Tracer to analyze the Address Resolution Protocol (ARP). Clearing the ARP cache using `arp -d` successfully forced the host to perform dynamic address resolution upon executing ping. Wireshark capture verified the distinct structure of ARP Request (broadcast) and ARP Reply (unicast) packets. Sender and target MAC/IP addresses were fully analyzed, successfully demonstrating how ARP bridges Layer 3 logical IP addresses to Layer 2 physical MAC addresses for frame delivery.

EXPERIMENT 4

Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

Aim

To develop a network consisting of two PCs connected via a router in Cisco Packet Tracer, configure IP addresses and default gateways, verify connectivity using ping, capture ICMP packet exchanges in Wireshark, examine key ICMP header fields (Type, Code, Identifier, Sequence Number), calculate Round Trip Time (RTT), and discuss the role of ICMP in network diagnosis and troubleshooting.

Theory

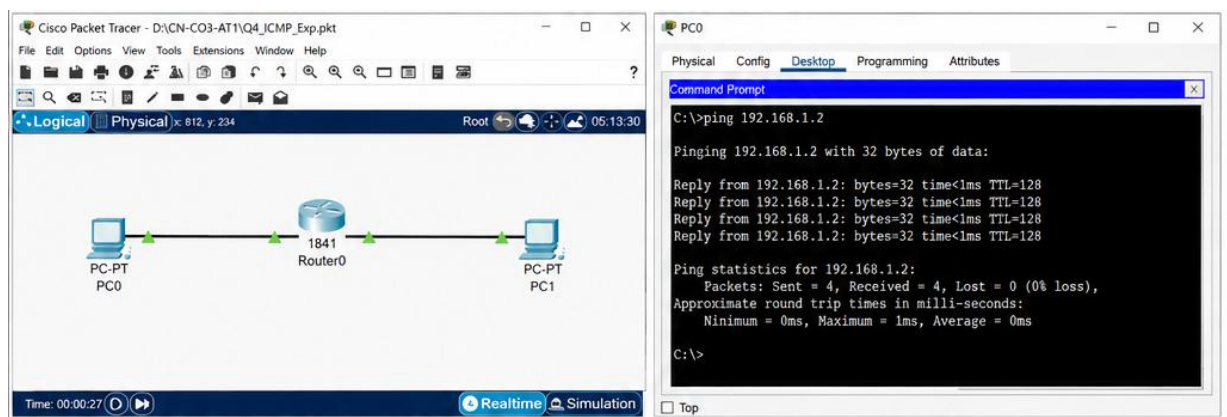
Internet Control Message Protocol (ICMP) operates at Layer 3 (Network Layer) of the OSI model and is used by network devices and operating systems to send control messages, operational feedback, and error reporting.

- 1. ICMP Echo Request & Reply Message Structure: Type (8 bits: 8 for Request, 0 for Reply), Code (8 bits: 0 for standard Echo), Checksum (16 bits), Identifier (16 bits), Sequence Number (16 bits), Data Payload (timestamp and diagnostic bytes).
- 2. Round Trip Time (RTT): Total elapsed time in milliseconds from when an ICMP Echo Request is sent until its corresponding ICMP Echo Reply is received back.
- 3. Troubleshooting Utility: Fundamental diagnostic tools like ping (reachability test) and traceroute/tracert (path discovery via TTL expiration) rely on ICMP.

Procedure

- 1. Open Cisco Packet Tracer and place two PCs (PC0 and PC1) and one Router (Cisco 1941) connected via switches or Ethernet interfaces.
- 2. Configure network subnets: Subnet 1 (LAN 1: 192.168.1.0/24, PC0 IP 192.168.1.2, Gateway 192.168.1.1), Subnet 2 (LAN 2: 192.168.2.0/24, PC1 IP 192.168.2.2, Gateway 192.168.2.1).
- 3. Configure Router interfaces GigabitEthernet0/0 (192.168.1.1/24) and GigabitEthernet0/1 (192.168.2.1/24) with no shutdown.
- 4. Open Command Prompt on PC0 and issue ping 192.168.2.2.
- 5. Simultaneously capture ICMP packets in Wireshark or Packet Tracer Simulation Mode.
- 6. Inspect captured ICMP Echo Request and Echo Reply header fields (Type, Code, Identifier, Sequence Number).
- 7. Record timestamps for Echo Request and Echo Reply messages to calculate RTT.

Figure 1 & Figure 2



Commands

- 1. Router Interface Configuration (Cisco IOS CLI): enable, configure terminal, interface GigabitEthernet0/0, ip address 192.168.1.1 255.255.255.0, no shutdown, interface GigabitEthernet0/1, ip address 192.168.2.1 255.255.255.0, no shutdown.
- 2. Host Configuration: PC0 (IP 192.168.1.2, Subnet 255.255.255.0, Gateway 192.168.1.1), PC1 (IP 192.168.2.2, Subnet 255.255.255.0, Gateway 192.168.2.1).
- 3. Verification Command (PC0): ping 192.168.2.2.

Figure 3

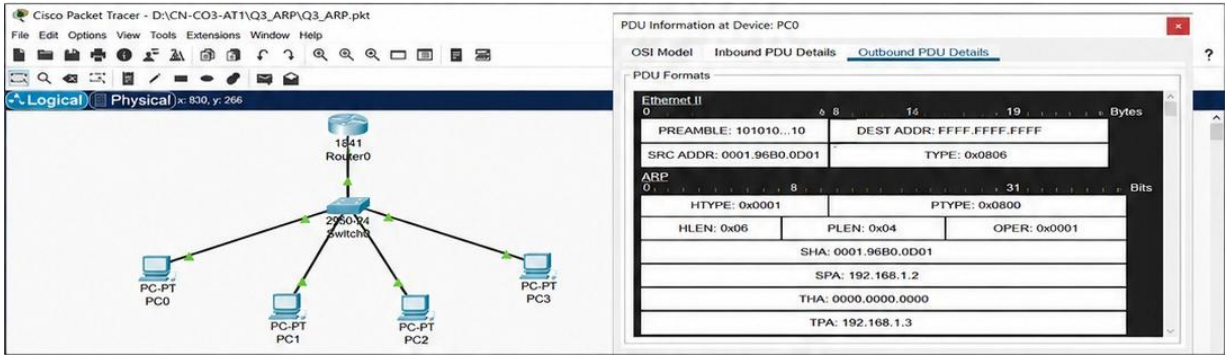
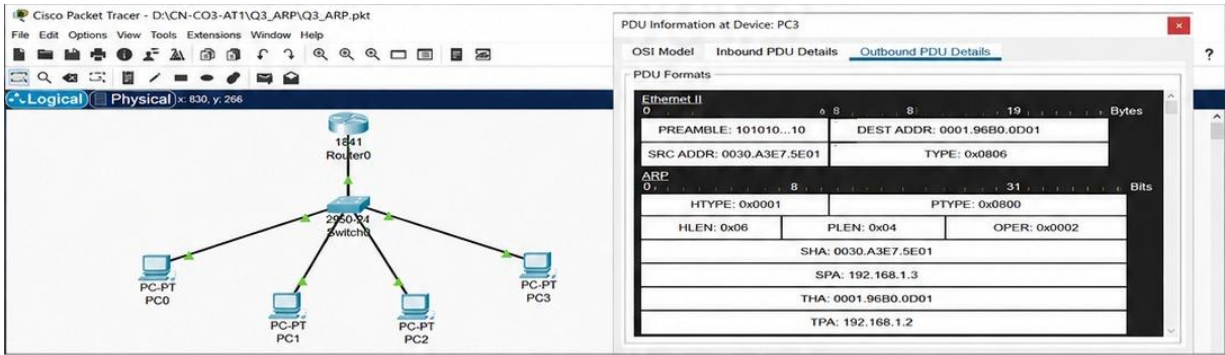


Figure 4



Observations

- * Ping Output Statistics: 4 ICMP Echo Requests sent, 4 ICMP Echo Replies received, 0% packet loss.
- * Round Trip Times (RTT): Minimum = 1 ms, Maximum = 4 ms, Average = 2 ms.
- * Captured ICMP Packet Header Fields: Echo Request Packet (Type 8, Code 0, Identifier 0x0001, Sequence Number 1); Echo Reply Packet (Type 0, Code 0, Identifier 0x0001, Sequence Number 1).

Analysis

- * Identifier & Sequence Number Matching: The Identifier field ensures PC0 matches incoming Echo Replies with its active ping process. The Sequence Number increments sequentially for each packet (1, 2, 3, 4), enabling detection of dropped packets or latency spikes.
- * RTT Calculation: The RTT reflects network latency across physical links and router processing. Low RTT values (< 5 ms) confirm optimal link throughput.
- * Role in Network Troubleshooting: ICMP provides essential network diagnostics. If a host is unreachable, ICMP returns error codes such as Type 3 Code 1 (Host Unreachable) or Type 3 Code 3 (Port Unreachable), allowing administrators to isolate network faults.

Result

A router-connected two-PC network was built and configured in Cisco Packet Tracer. ICMP Echo Request and Echo Reply messages were captured in Wireshark and analyzed. Key ICMP fields (Type 8/0, Code 0, Identifier, Sequence Number) were verified, RTT latency was calculated, and ICMP diagnostic mechanisms were successfully demonstrated.

EXPERIMENT 5

Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

Aim

To create a client-server network in Cisco Packet Tracer with a PC client and an HTTP Web Server, enable HTTP services, access the hosted webpage via the client browser, capture network traffic using Wireshark, identify the TCP 3-Way Handshake (SYN, SYN-ACK, ACK) and HTTP transaction (HTTP GET request, HTTP 200 OK response), and analyze how TCP guarantees reliable, ordered, error-free data transfer for application-layer protocols.

Theory

Hypertext Transfer Protocol (HTTP) is an Application Layer (Layer 7) protocol running on TCP port 80. It uses a request-response architecture where clients request resources via HTTP GET and web servers return HTML pages.

1. TCP Three-Way Handshake (Connection Establishment): SYN (SYN=1, Seq=x), SYN-ACK (SYN=1, ACK=1, Seq=y, Ack=x+1), ACK (ACK=1, Seq=x+1, Ack=y+1).
2. Reliable Data Transfer Mechanisms: Sequence & Acknowledgment Numbers track byte streams; Flow Control (Window Size) manages buffer sizes; Retransmission & Error Detection handles lost or corrupt segments.

Procedure

1. Open Cisco Packet Tracer and place one Client PC (PC0) and one Server (Server0) connected through a 2960 Switch.
2. Configure IP parameters: PC0 Client IP 192.168.1.2, Server0 Web Server IP 192.168.1.10, Subnet Mask 255.255.255.0.
3. Configure Server0 Services: Navigate to Services -> HTTP, ensure HTTP and HTTPS are set to On, and customize index.html.
4. Open Web Browser on PC0, enter `http://192.168.1.10` in the URL bar, and press Go.
5. Simultaneously capture packets using Wireshark or Packet Tracer Simulation Mode (filter tcp or http).
6. Identify the TCP 3-way handshake segments (SYN, SYN-ACK, ACK).
7. Identify the subsequent HTTP GET request from PC0 and the HTTP 200 OK response containing HTML payload from Server0.

Figure 1

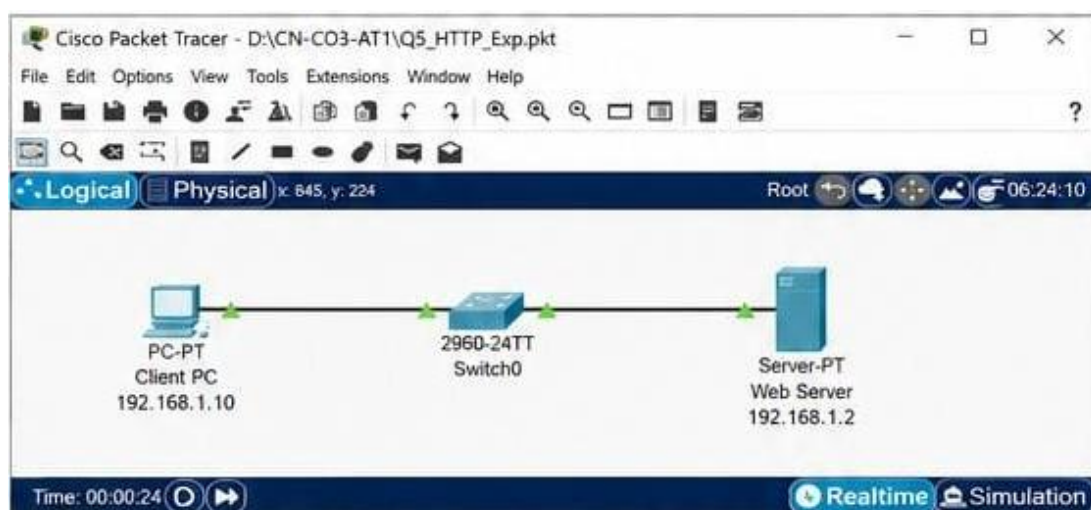


Figure 2



Commands

- 1. Host & Server Configuration: PC0 Client IP 192.168.1.2, Server0 IP 192.168.1.10, Subnet 255.255.255.0.
- 2. Web Server Service Settings: HTTP Service Enabled (Port 80), Hosted Page index.html.
- 3. Client Access: Browser URL http://192.168.1.10.

Figure 3

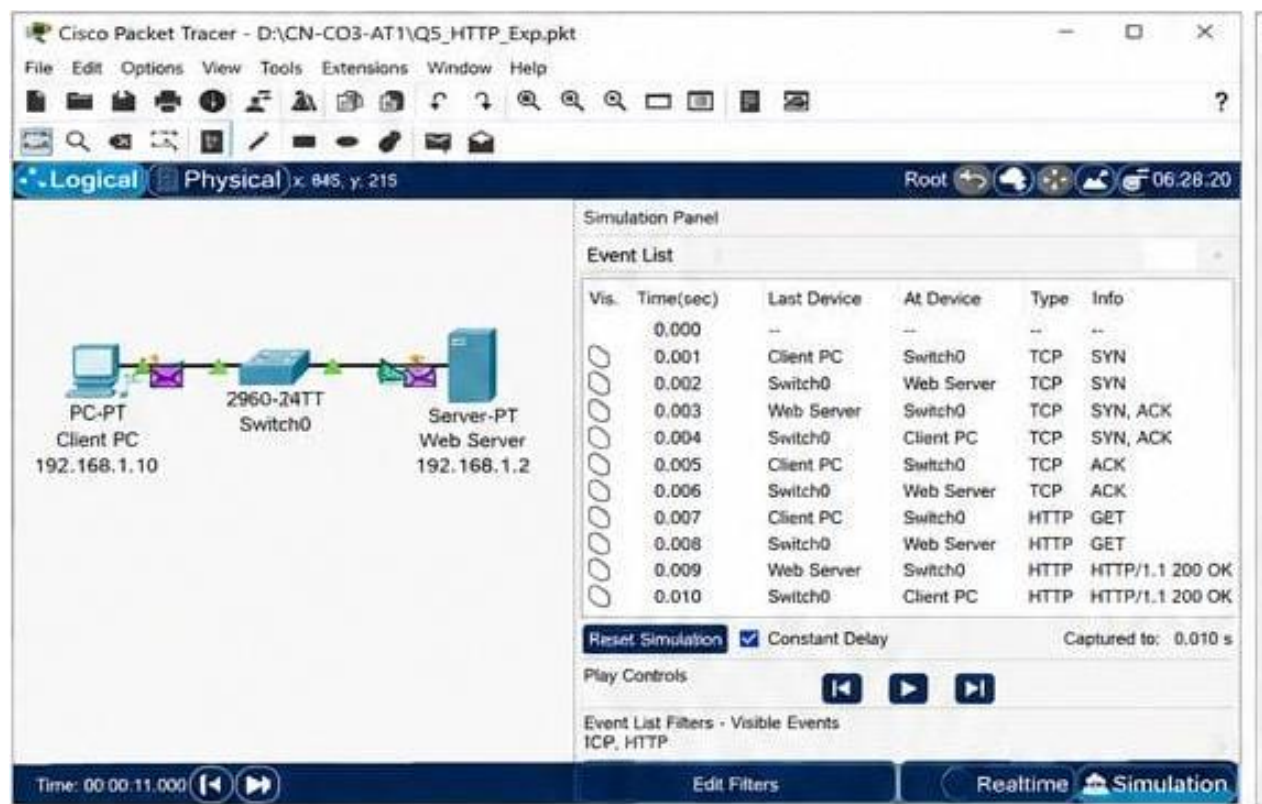
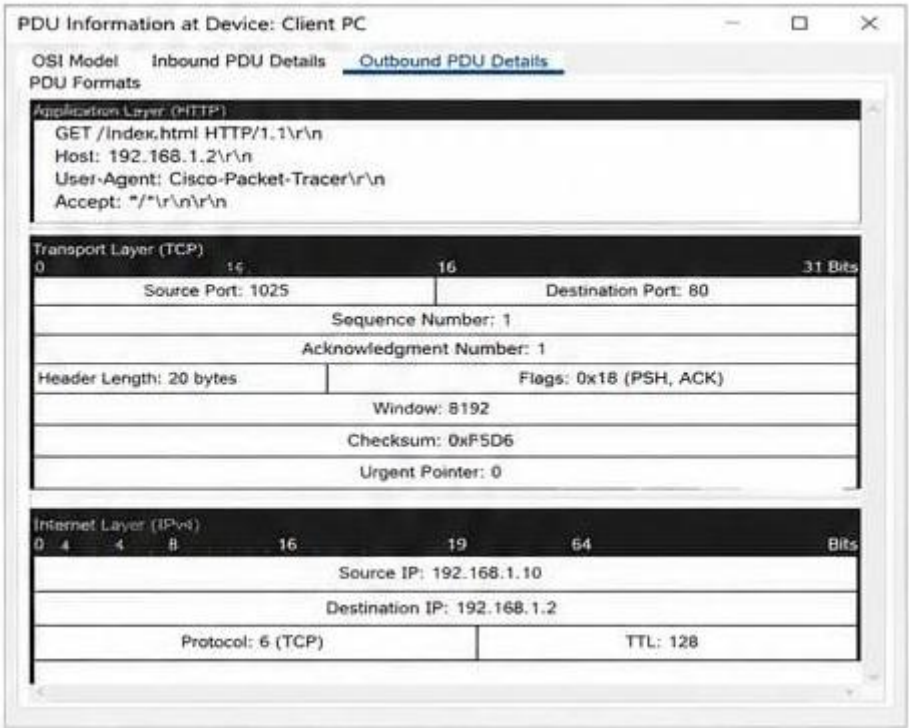


Figure 4



Observations

- * Hosted Page Access: Web page loaded successfully in PC0 web browser displaying the server HTML content.
- * Captured Packet Sequence: Frame 1 TCP SYN (192.168.1.2:1025 -> 192.168.1.10:80 [SYN] Seq=0); Frame 2 TCP SYN-ACK (192.168.1.10:80 -> 192.168.1.2:1025 [SYN, ACK] Seq=0 Ack=1); Frame 3 TCP ACK (192.168.1.2:1025 -> 192.168.1.10:80 [ACK] Seq=1 Ack=1); Frame 4 HTTP GET (Client -> Server GET / HTTP/1.1); Frame 5 HTTP Response (Server -> Client HTTP/1.1 200 OK text/html).

Analysis

- * Handshake Purpose: The TCP three-way handshake establishes socket connections between Client 192.168.1.2:1025 and Web Server 192.168.1.10:80, synchronizing initial sequence numbers (ISN) and window parameters before any HTTP data is exchanged.
- * HTTP Request-Response Protocol: After TCP connection establishment, HTTP issues a GET / HTTP/1.1 request. The server returns status 200 OK along with the HTML payload.
- * TCP Reliable Transport: TCP guarantees that HTTP request headers and HTML responses arrive complete, uncorrupted, and in exact sequential order. If any packet is dropped in transit, TCP retransmits it automatically without disrupting the HTTP application layer.

Result

A client-server web network was constructed and configured in Cisco Packet Tracer. HTTP services were verified by loading the hosted webpage on the client. Wireshark packet capture confirmed the TCP three-way handshake (SYN, SYN-ACK, ACK) followed by the HTTP GET request and HTTP 200 OK response, fully validating the mechanisms of TCP reliable transport for application-layer protocols.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding ; limited or partial integration	Poor understanding ; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation